

1. Explain Each Steps of Array Insertion Thoroughly.

You are given the following array:

Array:

A = [12, 25, 8, 19, 33, 65]

Task:

1. Insert the value **20** at position **3** (1-based index).
2. Show each step of shifting elements.
3. Write the final array after insertion.
4. Clearly explain how the array is updated at each step.

2. Binary Search (Step-by-Step Explanation)

You are given a sorted array:

Array:

A = [3, 7, 12, 18, 25, 31, 40]

Task:

1. Perform binary search to find the element **25**.
2. Show each step clearly:
 - Low index
 - High index
 - Mid calculation
 - Comparison result
3. Write each iteration in a table format or step format.
4. Clearly mention how the search space reduces at each step.

3. Algorithm Writing

Write algorithms for the following problems:

a. GCD of Two Numbers

- Write a clear step-by-step algorithm.
- You may use Euclidean algorithm method.
- Clearly mention input, process, and output.

b. Second Largest Element in an Array

- Write a complete algorithm without sorting.
- Your algorithm should handle duplicate values.
- Clearly mention initialization and traversal steps.